

ASSIGNMENT – 06

Q6. Analyze the different communication mechanisms used in embedded systems by comparing serial and parallel communication, analog and digital communication, and synchronous and asynchronous protocols. Discuss how these categories differ in terms of data transmission, efficiency, complexity, and reliability, and evaluate their role in enabling effective interfacing between microcontrollers and external devices in real-world applications.

Serial Communication vs Parallel Communication:

Feature	Serial communication	Parallel communication
Data transmission	Bits are sent in a single wire or sequentially.	Here multiple bits send simultaneously on multiple wires.
Efficiency	Highly efficiency for long distances.	High efficiency for short distances.
Complexity	Low physical complexity.	High physical complexity
Reliability	high	low

Analog communication vs digital communication:

Feature	Serial communication	Parallel communication
Data transmission	Uses continuous signals.	Uses discrete values (0,1)
Efficiency	Higher bandwidth utilization for specific signals.	Efficient for data storage and processing.
Complexity	Simple hardware/sensors but sensitive for circuit components.	Requires ADCs/DACs (Analog to digital converter/digital to analog converter).
Reliability	low	Highly reliable

Synchronous protocol vs Asynchronous protocol:

Feature	Serial communication	Parallel communication
Data transmission	Data is synced with a shared clock pulse.	Data uses start/stop bits (no use any clock)
Efficiency	High	Lower
Complexity	High (requires an extra clock).	Low (requires only data wires)
Reliability	High (devices are perfectly synced).	Moderate (Requires matching bud rate on both ends).

Real-World Application:

SPI and I2C (Synchronous) are used when high speed and precision are needed close to the MCU. UART (Asynchronous) is the go-to for simple, long-distance communication like PC-to-controller debugging.